Zoology Legends

ZOO102

Important MCQS for Final TERM Spring 2023

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Zoology Legends

MCQS Spring 2023

- In animals, which organ maintains the volume of water?_**Kidney**
- In a sequence of hierarchy in which the female comes first, followed by the male, it is called: **Matrilineal/ protogynous**
- Amino acids are linked through: **Peptide bond**
- The main function of animals is to maintain: **Homeostasis**
- The process by which organic sulfur is converted into inorganic form i.e., hydrogen sulfide, elemental sulfur and sulfide minerals is called **mineralization**
- The step at which sulfides and elemental sulfur are converted to sulfates is called **oxidation**
- A **dependent variable** is the parameter used to determine the outcome of an experiment
- monosaccride form together… **Polysaccharides**
- These are the animals which retain eggs into the body but the young one gets all its food from yolk of egg. …. **Ovoviparous**
- Ammonia is formed during deamination of?_**Amino Acid**
- Who enzyme is essential for vertebrate skeleton and teeth?_ **Phosphorus**
- Outer layer of cytoplasm is called?__ **Ectoplasm**
- Name the simple sugar that are not further hydrolyzed? __ **Monosaccharides**

Zoology Legends

- Which is essential component of DNA, RNA, phospholipids?__ **Phosphorus**
- In mating process male introduces sperms into female body, sperms are motile they approach to female gametes within the body of female. Many sperms are packed into a ball like structure called **spermatophore**
- During first five months of development many small follicles are formed in the ovary of female fetus. These are called **primordial follicles.**
- In human the entire process of **spermatogenesis** and its transportation takes about **3 months**
- The total gestation period in human is **280 days**. This duration is divided into 3 months periods called **trimesters**.
- **Heart beat** can be heard by the end of **fourth month (18 Week).**
- Write the name of procedure that occure aftr the 4 month of pregnancy?__ **Heart beat**
- Write the name of reproduction that produce new offspring?__ **Sexual Reproduction**
- **Sexual Reproduction** It the types of reproduction in which new offspring are produced by the fusion of gametes
- Protostomes is an example of?__ **Spiral Cleavage**
- Biological science that addresses the questions of proximate causes is called **experimental science.** Which is break into two and more picese of fragments?__ **Fragmentation**

Zoology Legends

- **Fragmentation** is the process of asexual reproduction in which an animal breaks into two or more pieces
- The process in which one organism divides into two organisms is called **binary fission**
- **Ecology**_It is the branch of biology that deals with the relationship of organisms to one another and with their physical environment
- Organisms require energy in one form or the other.___ **Energy Flow**
- **Symbiosis** is a close and long term association between two different organisms
- Organisms are entirely dependent on symbiosis are obligate. E.g., **Fungus of lichen**
- **Viviparous** __These are the animals in which eggs develop into the body and the young one gets its nourishment from mother.
- Eggs with moderate amount of yolk are called **mesolecithal** eggs. e.g., lung fish and amphibians. While eggs with large amount of yolk are called **macrolecithal** eggs. e.g., reptiles and birds.
- Seminiferous tubule join a network of tubules called **rete testis**.
- cleavage occur in embryo of nematodes___ **Holoblastic Cleavage**
- **Holoblastic Cleavage:** The pattern of cleavage in which egg is completely divided into two cells
- "Two monosaccharides combine together to form a **disaccharide**
- Melting point of Palmitic acid… **62.9**

Zoology Legends

- **Histology**: Study of organisms at tissue level is called histology.
- **Entomology**: Entomology is a branch of zoology which deals with the study of insects.
- **Renewable resources** are those resources which are constantly available
- **Animal diversity** is the number of different types of animals living on this earth.
- An **independent variable** is parameter being tested in the experiment.
- A **dependent variable** is the parameter used to determine the outcome of an experiment while a **control variable** is that which can influence the results so it is kept constant.
- In chaparral annual precipitation.. **30-75cm**
- **Atmosphere** is the gaseous content of biosphere
- The final community during an ecological succession is called **climax community**.
- **Camouflage** It is the phenomenon in which the colour pattern of animal helps to hide the animal from others
- **Hibernation**: Time of low body temperature and decreased metabolism from weeks to months
- Eggs containing negligible amount of yolk are called **alecithal egg**. e.g., placental mammals. Amphibians have...eggs__ **mesolecithal eggs**
- **Radial cleave** is the characteristic of deuterostomes.

Zoology Legends

- **Meroblastic Cleavage:** The pattern of cleavage in which egg is not completely divided into two cells
- **Holoblastic Cleavage:** The pattern of cleavage in which egg is completely divided into two cells.
- **Parasitism**: It is a relationship in which on organism (parasite) live in or on other organism (host). Parasite gets benefit while host is harmed.
- The cells developing embryo and placenta release **human chorionic gonadotropin (hCG)**
- The tiny chorionic villi sent out from **blastocyst** form placenta
- Human have placenta formed form **chorion and allantois**
- Which is water soluble _ **Globular Protein**
- RNA molecules in nature work as enzymes are called **ribozymes**.
- The non-protein organic molecules require for enzyme activity is called **coenzyme**. e.g., vitamins
- An enzyme without its cofactor or coenzyme is called **apoenzyme**. Apoenzyme is catalytically inactive.
- An enzyme with its attached cofactor or coenzyme is called **holoenzyme**
- **Coniferous forest** are dominated by cone bearing, needle shape evergreen plants called **conifers**
- **Coniferous forests** are present in Chitral, Swat, Dir, Abbot Abad, Hilly Areas of Balochistan, Murree and Azad Kashmir

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- Which is essential component of DNA, RNA , phospholipids , vertebrates__**phosphorus**
- Biomass is _
- At an average there is **10%** biomass conversion efficiency at each trophic level.
- It is the efficiency by which animal at a trophic level convert food into new **biomass**
- **Pyramid of Biomass:** It is the representation of total mass of living matter present at each trophic level.
- The amount of energy stored in the form of biomass is called **net productivity**.
- **Logistic Population Growth:** S-shaped curve: At carrying capacity population growth levels off
- **Exponential Growth**: Population growth in which number of individuals increase by a constant multiple in each generation
- **Polysaccharides:** "When many monosaccharides join together through glycosidic bod, they form a polysaccharide"
- Outer layer of cytoplasm is called _ **Ectoplasm**
- Parthenogenesis is a process of production of an organism from an **unfertilized egg**.
- ***Ectosymbiosis***: It is a symbiosis in which symbionts live on the body surface of the host or in the digestive tact. E.g., Lice, barnacles and cleaner fish

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- **Endosymbiosis**: It is the relationship in which one symbiont live within the tissue of other. E.g., nitrogen fixing bacteria in root nodules, algae inside the corals
- How much NADH are yields during glycolysis___**Two**
- Marine water typically contain…Percent salt__**97%**
- Fresh water contains salt concentration about ___**3%**
- Less than **1%** of water is available to human.
- In temperate deciduous forest temperature varies from__-**30oC to 30oC.**
- Following stages of premature birth__ **Third Trimester**
- **Neurulation**: The process of forming neural tube is called neurulation.
- **Phagocytosis**: It is the uptake of solid materials by the cell.
- **Pinocytosis**: It is the uptake of liquid from extracellular environment.
- One gram of proteins give approximately 4kcal energy
- Shelled eggs with extraembryonic layers in reptiles__ **amniotic eggs**
- **Reptiles** are true terrestrial vertebrates. They have shelled eggs with extraembryonic membranes.
- **The Follicular Phase**: Developing follicle produces estrogen
- **histology** is study at tissue level
- Average precipitation is very low approximately less than 25cm/year in **Tundra Ecosystem**

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- Kelps and algae are plants in **Neritic Ecosyste**
- **Columnar epithelial tissue** are single layer of elongated cells
- Beekeeping for the production of honey and wax is called **apiculture** while rearing of silkworm for production of silk is called **sericulture**.
- **Fundamental Properties of Life:** "The ability to reproduce is Life."
- Coniferous forest have annual temperature **10°C**
- **Taxonomy**: Brach of zoology that deals with naming and classification
- Egg takes **5 days** to reach uterus
- Amphibian's are_ **Viviparous**
- Amphibians mostly have **external fertilization** in water in few internal.
- Formation of sperm is called…
- The process of sperm production is called **spermatogenesis**
- **Gametogenesis** is the process of formation of gametes.
- Which of the following hormones is secreted from pituitary gland__ **Prolactin**
- 1 gram protein has….**4 kcal energy**
- Kelps and algae are **plants**
- About **78%** of precipitation occurs over oceans.
- **DESERT ECOSYSTEM:** It is a region which receives annual precipitation less than **25cm**

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- Beekeeping is....**apiculture**
- **Sexual reproduction** is consider as more advantageous than asexual reproduction.
- **Reproduction** is the process of producing new individuals
- Average heart beat for human is **72 times/min** or **100,000 times/day**. In the life of 70 years heart beats for **about 2.5 billion times**
- The recommended amount of energy for our daily life is **30%**
- Those sequential hermaphrodites which are female in first phase of life and male in second phase of life called **protogynous** (femle first) e.g., wrasses.
- Those animals which are male in first phase of life and female in second phase of life are called **protandrous** e.g., clone fish.
- Potential life span of American robin is **11 years**, however average life **is 1.4 year**
- It is the first community established in an area devoid of life is called **pioneer community**
- **Water** is essential for life
- Parasites living on the body surface of host are **ectoparasites**. E.g., lice, leeches
- Parasites living within the body of host are **endoparasites**. E.g., tape worm, liver fluke, plasmodium
- Food chain is important To understand the problems of **biomagnification**.

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- One part of resources is used by one species and second part by other species. This mechanism is called **niche diversification**.
- Some bacteria convert nitrates back to nitrogen is called **denitrification**.
- Current documented biodiversity of earth is more than **1.6 million**
- The total Animal diversity of life documented is about **1.74 million**
- More than **1.5 million** species of animals have been documented.
- **Physiology**: The study of the function of organisms and their body parts.
- Tropical rainforest have **50% of plants** and animals diversity with just **6% land**.
- The process of sperms production is called **spermatogenesis** while the process of eggs production is called **oogenesis**.
- In eutherian mammals embryo is implanted deeply, fetus get nutrition through placenta, gestation period **20 days to 19** months.
- Average size of Ostrich egg is **15 x 13** cm
- **Biotic Potential:** The capacity of a population to increase maximally is called biotic potential
- **Temperature**: It is the measure of heat present in a body

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- **Ectoderm**: It is the outer most layer of cells formed during gastrulation.
- **Endoderm** is the inner most primary germ layer formed during gastrulation
- **Mesoderm** is one of the primary germ layer formed during gastrulation. It is present between ectoderm and endoderm.
- In protostomes cleavage is **spiral** while in deuterostomes cleavage is **radial**.
- **Cleavage** refers to cell divisions that occur during embryonic development.
- Initially all the cells of embryo divide at the same time this is called **synchrony**.
- As development proceeds, all cells don't divide simultaneously. This is called **asynchrony**.
- **Holoblastic Cleavage**: The pattern of cleavage in which egg is completely divided into two cells.
- **Meroblastic Cleavage**: The pattern of cleavage in which egg is not completely divided into two cells. This is due to large amount of yolk
- **Spiral Cleavage**: It is pattern in which cells of animal pole are not exactly laying above the vegetal pole cells. Cleavage takes place at oblique angle forming a spiral pattern. Example: protostomes cleavage

Zoology Legends

- **Radial Cleavage**:It is pattern in which cells of animal pole are exactly laying above the vegetal pole cells. Upper tier of cells is exactly on the lower tier of cells. Example: deurostomes cleavage
- **Kingdom** is the highest rank of classification.
- **Placental mammals** egg is **smallest** in animals kingdom.
- **Mammalian** cleavage is **slowest** in animal kingdom
- Protostomes belong to group **protostomia** of animal kingdom.
- Deuterstomes belong to group **deuterostomia** of animal kingdom.
- **Anatomy**: It is the study of the structure of complete organisms and their body parts.
- **Physiology**: The study of the function of organisms and their body parts.
- **Cell biology (Cytology**): Study of the structure and function of cell and its components.
- **Molecular Biology:** Molecular biology is the study of structure and function of the macromolecules (e.g. proteins and nucleic acids, Lipids) essential to life.
- **Histology:** Study of organisms at tissue level is called histology.
- **Embryology:** Study of development of embryo from fertilized egg.
- **Genetics:** Study of the mechanisms of transmission of traits from parents to offspring.
- Simple form of carbohydrates (**Monosaccharides**)

Zoology Legends

- Study of the structure of complete organisms and their body part (**Anatomy**)
- Information about Environmental problem (**Data**)
- **Observation** is the collection of information about a particular problem
- The information obtained from observations is called **data**
- Example of Tertiary Structure (**Amylase**)
- Bee keeping is called (**apiculture**)
- Precipitation of Chaparral (**30-75cm)**
- In Protostomes coelom is formed by splitting of mesoderm. This pattern is called (**schizocoelous**)
- Biggest of all levels of Ecological hierarchy (**Biosphere**)
- Outer most layer of cells formed during gastrulation (**Ectoderm**)
- Ectodermal cells migrate towards blastopore and spread over entire embryo. This is called (**epiboly**)
- **Ecosystem:** It is the study of all population in a community and their physical environment
- **Hemimetabolous** metamorphosis (Incomplete metamorphosis) is the mode of development of certain insects like aphid, grasshopper and cockroach.
- Arctic tundra is present in northern **hemisphere**
- **Hibernation**: Time of low body temperature and decreased metabolism from weeks to months

Zoology Legends

- **Coexistence**: It is the phenomenon in which competing species live togetherPyrimed….
- **Carbon** is the basic element of all organic molecules.
- Melting point **increase** with increase in number of carbon atoms
- **Sprouters** are dormant meristem, storage and intact root help to regenerate quickly while **seeders** are long lived seeds, germination induced by fire
- **Lotic Ecosystem:** It is the aquatic ecosystem of flowing water. It includes brooks, streams and rivers
- **Lentic Ecosystem:** This is the ecosystem of standing water. Lentic ecosystem include lakes and ponds.
- Study of complete structure of organisms and body parts (**anatomy**)
- conclude something from general information (**deductive reasoning**)
- A **deductive reasoning** hypothesis is that conclude something from general information while
- **Inductive reasoning** hypothesis given some general conclusion from very specific information.
- carbohydrates that cannot be hydrolyzed (**monosaccharides**)
- ___is involved in cell recognition and determination of cell behavior (**Glycocalyx**)
- ____is the process that Force small molecules across selective permeable membrane using hydrostatic pressure (**filtration**)

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- Double membrane organelle involved in energy production (**mitochondria**)
- Double membrane structure that separate nucleus from cytoplasm (**nuclear membrane**)
- Tissue present in lungs air sacs lining of heart and blood vessels (**simple squamous epithelial tissue**)
- The reaction which absorb energy (**androgenic reaction**)
- ___NADH are used during glycolysis (**2**)
- Longest gestation period is in (***elephant***)
- Which complete after fertilization (**meiosis 2**)
- Deamination of what produces ammonia (**amino acid**)
- Tropical rainforest have annual rainfall more then (**200cm**)
- The rigid material on the surface of earth (**lithosphere**)
- Group of different species (**community**)
- Phenomenon in which ecologically related species exert strong selective influence on other (**coevolution**)

ZOOLOGY
LEGENDS

Zoology Legends

Repeated Questions Final Spring 2023

Holoblastic Cleavage

The pattern of cleavage in which egg is completely divided into two cells. Egg contain little or moderate amount of yolk. Cleavage in fertilized eggs of Amphibians, mammals, echinoderms, flatworms and nematodes.

Monozygotic Twins

Monozygotic Twins are from single zygote and are identical twins. They are strikingly alike and of same sex. About 33% have separate placenta. Majority have separate amniotic sac. Small proportion have one placenta and one amniotic sac which indicate separation occur after 9 day of pregnancy. There is conjoining risk, the condition is Siamese twining.

Dizygotic Twins

Dizygotic Twins are produced from two zygotes. They are called as fraternal twins. They do not have striking similarities and may be of same or different sex. They have separate placenta and separate amniotic sac.

Protogynous & Protandrous

Sequential hermaphrodites have one sex in first phase of life and opposite sex in other phase of life. Means if animal

is male in first phase of life, it will be female in second half of life.

Those sequential hermaphrodites which are female in first phase of life and male in second phase of life called **protogynous** (femle first) e.g., wrasses.

Those animals which are male in first phase of life and female in second phase of life are called **protandrous** e.g., clone fish.

Decomposers

These are the organisms which obtain energy by breaking down dead organic matter into simpler compounds. They are also called **saprotrophs**. Chief decomposers of an ecosystem are bacteria and fungi. They release extracellular enzymes which breakdown substrate molecules.

Organisms at all trophic level when die are used by decomposers. If there are no decomposers mounts of dead bodies will be everywhere

Isotonic Solution:

When solute concentration is same outside and inside of the cells. Water molecules move inside and out of the cell at same rate.

Hypertonic Solution:

When solute concentration is higher outside the cells than inside. Water molecules move out and the cell shrinks. This condition is called crenation.

Hypotonic Solution:

When solute concentration is lower outside the cell than inside. Water molecules move into the cell which causes the cell to swell and burst.

Reproduction & its Importance

Reproduction is the process of producing new individuals. It is the basic feature of all forms of life. It is essential for existence of species and also for continuity of life on earth. Capability to reproduce is called reproductive potential. All the animals have great reproductive potential. If resource are unlimited a single species of fishes may fill the all oceans.

Animals reproduce by two basic means: Asexual reproduction and sexual reproduction

Types of Nucleotide

On the basis of sugar

- Deoxyribonucleotide
- Ribonucleotides

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On the basis of nitrogen base

- Adenine
- Guanine
- Cytosine
- Thymine/Uracil

Define fragmentation

Fragmentation is the process of asexual reproduction in which an animal breaks into two or more pieces and each piece is capable of growing into a new organism. Fragmentation is common in different groups of animals including cnidarians, Platyhelminthes and echinoderms

Difference b/t Sprouters and Seeders

There are two types of plants: **sprouters and seeders**.

Sprouters are dormant meristem, storage and intact root help to regenerate quickly while **seeders** are long lived seeds, germination induced by fire.

Plants include toyon, chamise, poison oak, scrub oak, yucca and sumac. Important **animals** are insects, rodents, rabbits, lizards, snakes and mule deer.

Define dependent and independent variables

Parameters in experiment that may change is called Variables. A hypothesis is tested by experiments. An experiment has three set of variables.

An independent variable is parameter being tested in the experiment.

A dependent variable is the parameter used to determine the outcome of an experiment while a control variable is that which can influence the results so it is kept constant.

For example, an experiment was set to measure the effect of day light length on reproductive readiness of mice. Here, daylight length is independent variable, frequency of mating is dependent variable while temperature is a control variable

Explain Rough & Smooth endoplasmic reticulum

There are two types of endoplasmic reticulum: Rough endoplasmic reticulum (RER) and Smooth endoplasmic reticulum (SER). Endoplasmic Reticulum with attached ribosomes on their surface are called **rough endoplasmic reticulum** (RER). Presence of ribosomes give them rough appearance.

While endoplasmic reticulum without ribosomes on their surface are called **smooth endoplasmic reticulum**.

Where does coexistence occur?

Coexistence

It is the phenomenon in which competing species live together

How Coexistence Occurs?

Coexistence can occur when species utilize resources in slightly different way. MacArthur studied 5 species of warblers that use same caterpillar from different region of trees

Budding

Asexual reproduction in which an outgrowth appears on the body of animal which grows into complete organism. e.g., hydra, obelia and sponges.

In budding, a group of cells rapidly divides and form a small outgrowth called bud. This bud increases rapidly and form a complete organism. The bud may get detach from the parent or sometimes it may remain attached with the parental organism and form a closely associated group of individuals called a colony. Members of a colony belong to same species.

In sponges internal buds are produces which are called as gemmules. Gemmule is a collection of cells surrounded by body wall. When parent dies, each gemmule gives rise to a new individual.

Lentic Ecosystem

This is the ecosystem of standing water. Lentic ecosystem include lakes and ponds. Vertical stratification is common especially in temperate region. In summer, upper layer is warm while lower layer is cold and more dense, so no

mixing. In fall, upper layer is dense so mixing of water occurs.

Photosynthesis mainly in upper layer of water as light is filtered by water. Phytoplankton are main producers of lentic ecosystem.

Plant life include phytoplankton, periphyton algae, emergents, floating- leaves, submersed and free floating plants. Animal life include zooplankton, invertebrates like crabs, crayfish and shrimps, molluscs while vertebrates mainly includes variety of fishes, reptiles and birds.

Lotic Ecosystem

It is the aquatic ecosystem of flowing water. It includes brooks, streams and rivers. Attached aquatic plants and algae provide some of the primary productivity. Most of the primary production comes from the surrounding plants by falling their materials during autumn and rainfall. Lotic systems typically have high oxygen level.

Turbulence of water while flowing over rocks, logs and other objects causes oxygenation. This support the animal life in lotic system. Animals include different aquatic invertebrates, larvae of insects, fishes, amphibians reptiles birds and mammals.

Kingdom

Kingdom is the highest rank of classification. All the living organisms are distributed into five kingdoms i.e., monera,

protista, fungi, plantae and animalia. All the animals are included in kingdom Animalia.

Microtubules

Microtubules are hollow, slender, cylindrical structures. They are largest cytoskeletal elements. Each microtubule is about 25nm in diameter and made up of tubulin protein. Microtubules help in movement of organelles and chromosomes.

They are part of transport system (e.g., secretory vesicle movement) in the cell and are involved in overall shape changes of cell during specialization. Microtubules make different structures like mitotic apparatus, cilia, flagella and centrioles.

Disadvantage of sexual reproduction

Sexual reproduction also has certain disadvantages, some of which are as follows:

1) Asexual animals: sexual reproduction is not feasible for animal which rely on asexual means only, especially population with one sex only.
2) Adaptive genes combination dismantle: In sexual reproduction new genetic combination arise. This will upset the already established adaptive gene combination.

3) Wastage of energy: in sexual reproduction a large number of gametes are formed. Many of which are not fertilized. This will cause a significant loss of energy.

Oceanic ecosystem

Oceanic Ecosystem starts from continental shelf. It is the open sea. As it is nutrient poor so, it is relatively unproductive. Virtually all primary productivity is in the upper water. Phytoplankton in the upper water are the main producers of oceanic system. Invertebrate animals include worms, echinoderms and molluscs. Vertebrates include fishes from small fishes to whales and sharks.

At depth no light so the animals here have luminescent organs which serve as torch light. These animals depend on nutrient coming from upper layers.

Evolutionary and experimental science

Biological science that addresses the questions of proximate causes is called experimental science. It proceeds through experiments. E.g., process of maintaining body temperature, process of cell division.

Biological science that addresses the questions of ultimate causes is called evolutionary science. It proceeds through comparative methods E.g., comparison of hemoglobin structure of human and fish.

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Define Isomerism. Give suitable example of carbohydrates

- Molecules with same molecular formula but different structural formula
- All disaccharides are isomers of each other.
- Molecular formula C12H22O11

Write three functions of epididymis of male reproductive system

Epididymis: Each testes has a large number of seminiferous tubules. These are the sites for sperm production. Seminiferous tubule join a network of tubules called rete testis. After that there is a dilated tubule called epididymis.

Epididymis has three functions:

1) Storage and maturation of sperms.
2) Smooth muscles that help to propel sperm during ejaculation and
3) Transfer sperms to duct deferens.

Describe Coral Reefs

Coral Reef is the underwater ecosystem. They are formed by colonies of stony corals.

- Coral are members of class Anthozoa.
- They grow in association of zooxanthellae.

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- Polyps of corals are held together by calcium carbonate.
- Reefs are formed over thousands of years in warm, clear and relatively shallow waters.
- The greatest depth for their growth is 50-60 m.
- Below this light is insufficient to support zooxanthellae.
- Coral Reefs are the richest source of biodiversity in oceans.
- Coral reefs occupy less than 0.1% of oceans, yet they are home of more than 25% marine species.
- Residents include sponges, cnidarians, nematodes, crustaceans, molluscs, echinoderms, tunicates and fishes.

What do we study in zoology?

Animal life is very fascinating to human being. It includes a great variety of animals ranging from microscopic invertebrates to very large size vertebrates. Beautiful domestic pets, gorgeous wildlife and stunning aquatic animals are all studied under the heading of zoology.

When we see living organisms different questions rise in our mind like;

How and where they live?

What they eat?

What they do?

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Describe Intertidal Zone

The Intertidal Zone extends from splash zone ocean to low tide marks. This region is subjected to the action of both tides and waves. It is the harshest and the richest of all marine environments. Here is pounding surf, sun, wind, rain, erosion, sedimentation and temperature fluctuation. This zone has bounty of habitat and plenty of nutrients. Here, plants are larger.

They include seagrass, red algae and mangroves. Motile animals like fishes and crabs move with tides. Other such as amphipods, isopods, ghost crabs and polychaetes burrow in the sand and mud.

Water trapped in tidal pools support a rich animal life such as echinoderms, barnacles, limpets and chitons. Attached animals close shells to avoid drought during low tides.

Morula cleavage / Morulation

Morulation: Cleavage results a ball of cells called morula. Morula is a solid ball of cells without any cavity. Cell towards animal pole are smaller while larger at vegetal pole due to presence of yolk.

Difference between loose and fibrous connective tissue

There are two general types of tissues: Loose connective tissues and hard connective tissues.

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Loose connective tissues are strong flexible fibers of collagen which are interwoven with elastic fibers. They are located in epithelia of skin and throughout body. They wrap and cushion organs.

In **fibrous connective tissues** collagen fibers densely packed, parallel. They are located in dermis of skin; submucosa of digestive tract; fibrous capsule of organ and joints. Their major function is to provide structural strength.

Difference between radial and spiral cleavage

Spiral Cleavage:

It is pattern in which cells of animal pole are not exactly laying above the vegetal pole cells. Cleavage takes place at oblique angle forming a spiral pattern.

Example: protostomes cleavage

Radial Cleavage:

It is pattern in which cells of animal pole are exactly laying above the vegetal pole cells. Upper tier of cells is exactly on the lower tier of cells.

Example: deurostomes cleavage

Importance of nucleotides

Nucleotides are organic molecules that play several important roles in biological systems:

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- Nucleotides are the building blocks of nucleic acids, including DNA and RNA. They are composed of a nitrogenous base, a pentose sugar, and a phosphate group. Nucleotides combine to form the genetic code, storing and transmitting genetic information in living organisms. They are essential for the replication, transcription, and translation processes involved in protein synthesis.
- Nucleotides provide energy in the form of nucleoside triphosphates (such as ATP, GTP, CTP, and UTP). These high-energy molecules are used throughout the cell to fuel various cellular processes, including metabolism, synthesis of proteins, cell division, and movement of cellular components.
- Nucleotides are involved in cell signaling. For example, cyclic guanosine monophosphate (cGMP) and cyclic adeosine monophosphate (cAMP) are nucleotides that function as secondary messengers, relaying signals within cells and regulating various cellular processes.
- Nucleotides serve as cofactors for enzymatic reactions. They are incorporated into important coenzymes such as coenzyme A, FAD, FMN, NAD, and NADP+, which are essential for various metabolic pathways and enzymatic reactions in the body.
- Nucleotides have a role in immune function. They contribute to the development and activation of immune

cells, including natural killer cells, which play a crucial role in defending against infections and diseases.

- Nucleotides can have effects on memory and cognitive function. Dietary nucleotides have been found to improve memory and cognitive abilities in some studies.

Difference between synchrony and asynchrony

Cleavage refers to cell divisions that occur during embryonic development. Initially all the cells of embryo divide at the same time this is called synchrony. As development proceeds, all cells don't divide simultaneously. This is called asynchrony. Eggs with evenly distributed yolk have cleavage pattern that result uniform size blastomeres. Egg with unevenly distributed yolk result in unequal size blastomeres.

Mineralization and oxidation in sulfur cycle (3 marks)

The process by which organic sulfur is converted into inorganic form i.e., hydrogen sulfide, elemental sulfur and sulfide minerals is called **mineralization** and the step at which sulfides and elemental sulfur are converted to sulfates is called oxidation. Sulfates are reduced by plants, fungi and bacteria and incorporated

Difference between renewable and non-renewable resources (5 marks).

Natural resource can be divided into two main types: Renewable resources and Non-renewable resources.

Zoology Legends

Renewable resources are those resources which are constantly available. After using they are reasonably replaced or recovered e.g., water, sunlight, air and biomass.

Non-Renewable Resources are those which once exhausted are no more available. They are not replaced or recovered e.g., oil, gas and minerals

Importance of phosphorus (5 marks)

- Phosphorus is an essential component of DNA, RNA, ATP, Phospholipids, Vertebrate skeleton and Teeth.
- Sedimentary rocks contains deposits of phosphorus.
- Extraction of phosphorus takes place by weathering, rain and other erosion processes.
- Plant, microbes and fungi absorb phosphorus from soil.
- Animals use phosphorus by utilizing plants.
- Some phosphorus also comes through water.
- As plants and animals die, phosphorus is return back to the environment via decomposition.
- Living organisms may again use this phosphorus.
- Some phosphorus may run along with water and become part of sediments.
- Human has significant impact by using phosphorus in fertilizers.
- Agricultural runoff takes phosphorus to water bodies which excessive growth of algae. This phenomenon is called eutrophication.

Zoology Legends

Common terms related to reproduction

Fertilization: Fusion of egg and sperm, forming a zygote.

Gestation: Pregnancy period in mammals from conception to birth.

Puberty: Adolescence stage with sexual maturation.

Ovulation: Release of mature egg from ovary in females.

Embryo: Developing stage after fertilization, before birth/hatching.

Explain asexual reproduction

It is the type of reproduction in which offspring are produced without fertilization of gametes. In asexual reproduction the basic method of cell division is mitosis which involves no genetic recombination.

So, the animals produced by asexual reproduction are identical to their parents and siblings. Asexual reproduction is common the common method of reproduction in protozoa (Amoeba, paramecium and euglena) and lower invertebrates (hydra, jellyfish, and flatworms).

Inductive and deductive reasoning

On the basis data collected through observation and their previous knowledge scientists give a reasonable explanation to the natural question that is called a hypothesis.

Zoology Legends

A hypothesis may base on deductive reasoning or inductive reasoning.

A deductive reasoning hypothesis is that conclude something from general information while **the Inductive reasoning** hypothesis given some general conclusion from very specific information.

Density-independent Factors

These factors affect population regardless of their number

- Weather conditions
- Earth quake
- Pollutants
- Deforestation
- Hurricane
- Fire

Advantages of asexual reproduction

Advantages of Asexual Reproduction

1) **Species survival:** Asexual reproduction is specially the beneficial to those species have one sex only.
2) **Minimal energy requirements:** Only one sex is involved so less amount of energy is required.
3) **Positive genetic combinations passed on to next generation:** Because there is no genetic recombination, the good traits acquired of an organism are transmitted to next generation

4) **Rapid Increase in Population:** Under stable conditions animals reproduce asexually causing a rapid increase in population

Neritic ecosystem

The Neritic Ecosystem extends from littoral zone to the edge of continental shelf. It is also called the shallow water. Nutrients are delivered by rivers and upwelling at the edge of continental shelf. This region is more productive than open sea. Planktonic and attached algae contribute to the primary productivity.

Plants include kelps, algae, seaweeds, seagrasses and phytoplankton. Invertebrate animals include crabs, sponges, echinoderms, corals, and polychaetes whereas, vertebrate fauna include different types of fishes such as code, tuna, herring, capelin and many other species.

Structure of DNA

The structure of DNA is a double helix formed by two polynucleotide chains. Each chain consists of nucleotides, which are composed of a sugar molecule (deoxyribose), a phosphate group, and one of four nitrogen-containing nucleobases: adenine (A), cytosine (C), guanine (G), or thymine (T).

The two chains are held together by hydrogen bonds between the nucleobases. Adenine forms acomplementary base pair with thymine, and cytosine pairs with guanine.

Zoology Legends

Hierarchy of classification

The Hierarchy of classification of animals is as follow:

- Kingdom
- Phylum
- Class
- Order
- Family
- Genus
- Species

The smallest group of classification system is species

Oviparous and viviparous

a. Oviparous

These are the animals which lay eggs and development takes place outside the body of animals. Fertilization may be internal or external. Majority of invertebrates and many vertebrates are oviparous.

b. Ovoviparous

These are the animals which retain eggs into the body but the young one gets all its food from yolk of egg. Both fertilization and development is internal. Many members of annelids, brachiopods, gastropods, fishes and reptiles.

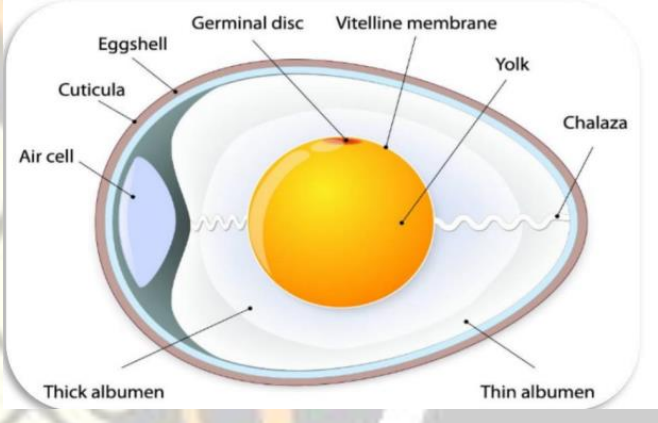
c. Viviparous

These are the animals in which eggs develop into the body and the young one gets its nourishment from mother. Both

fertilization and development is internal. Scorpions, most mammals, elasmobranch fishes, few amphibians and reptiles are viviparous.

Structure of Avian Egg

Membranes and fluids are collected around the egg while moving through oviduct. Water and albumin is deposited around the egg. This protect from desiccation and mechanical damage.



Albumin is used for protection and a source of nutrition. Two denser strands of albumin called chalaza are attached to the egg shell. They suspend egg in the center of albumen. Shell gland secrete $CaCO_3$ which deposit around the egg.

It is also impregnated with proteins. Thousands of tiny pores (40-50µm) are for exchange of gases. Inside the shell two membranes are present. At the round end of egg an air pocket is present between two membranes.

Air pocket enlarges with time. Near the time of hatching the chick breaks air pocket, the air inflates its lungs and the chick start breathing.

Fission

Fission (Lenten word fissio, the act of splitting) is a type of asexual reproduction in which one organisms splits into two or more organisms.

There are two basic types of fission: Binary fission and multiple fission. The process in which one organism divides into two organisms is called **binary fission**. Both are approximately equal size. Binary fission may be asymmetrical, transverse and longitudinal.

Asymmetric organisms like amoeba may divide into two from any side, such fission is called asymmetric fission. When a line of cleave transversely divide an organism into two is called transverse fission. e.g., binary fission in paramecium.

When an organism is divided longitudinally into two daughter organisms, this is called longitudinal fission. e.g., binary fission in planaria and euglena.

Multiple fission is a number of successive divisions of nucleus followed by cytoplasmic division in which one cell divides into many cells. e.g., amoeba, tapeworm. Amoeba form cyst under unfavorable conditions and form new amoebae by multiple fission.

Define intermediate filaments

Intermediate filaments are heterogeneous group of cytoskeletal fibers. They are made up of more than 70

different types of proteins. They are of intermediate size (10nm diameter). Intermediate filaments help to maintain cell shape, maintain spatial arrangement of organelles and promote mechanical activities in cytoplasm.

Theory of Chemical evolution / Theory of cell evolution

Life evolved about 4.0 billion year ago. **According to theory of chemical evolution** simple molecules were first converted into monomers of carbohydrates, proteins, lipids and nucleic acids. These monomers later joined to form polymers.

All these polymers were suspended in oceans giving up a soup like consistency. Polymers later arranged to form a primitive type of cell called Protocell. This was a simple self replicating system. It was a simple bacteria like primary heterotroph organism.

As food availability became limited, natural selection acted upon these simple cells and self-replicating systems were more successful. Cells began to evolve and genetic code was established and protein synthesis mechanism was developed.

Fossil record shows that life was existing about 3.8 billion years ago. Eukaryotes (membrane bounded nucleus and organelles) evolved about 1.5 billion years ago.

Nitrogen cycle

- Nitrogen fixation converts atmospheric nitrogen into forms usable by plants.
- Plants take up nitrogen compounds from soil for growth.
- Decomposition returns nitrogen to soil; nitrification converts it to nitrate.
- Denitrification releases nitrogen back into the atmosphere.
- Human activities like farming impact the nitrogen cycle through fertilizers.

Nucleotides

- Important biomolecules essential to life
- Perform different cellular activities
- They are also building blocks of nucleic acids

• Three components

- Five carbon/pentose sugar/ (ribose, deoxyribose)
- Phosphate group
- Nitrogen containing base

Characteristics of biome found in small patches / THE CHAPARRAL

It is the biome fount in small patches in most of the continents. It is characterized by very hot and dry summer with temperature up to 37.5oC. Winter is very mild and

usually about 10oC and most of the rain occur in winter. Annual precipitation is 30-75cm.

Soil is thin, rocky, nutrient poor, highly susceptible to erosion and contains low amount of moisture. Plants of chaparral are low and shrubby. They have tough and waxy leaves which are broad leaves and evergreen. Plants have no thick bark so not very resistant however their recovery is very fast.

There are two types of plants: sprouters and seeders. Sprouters are dormant meristem, storage and intact root help to regenerate quickly while seeders are long lived seeds, germination induced by fire. Plants include toyon, chamise, poison oak, scrub oak, yucca and sumac. Important animals are insects, rodents, rabbits, lizards, snakes and mule deer.

Chaparral is represented by west coast of United States and South America, South Africa, western tip of Australia and coastal area of Mediterranianean.

Effect of pesticides on children vs. adults

- Children's developing bodies are more sensitive to pesticides.
- Children consume more relative to their weight, leading to higher pesticide exposure.
- Pesticides can impact brain development and behavior in children.

- Developing hormonal systems make children susceptible to pesticide effects.
- Health issues from early-life exposure can have lifelong impacts.

Factors of abiotic stress methods

Biotic Factors

All living organisms in ecosystem make biotic factors.

- Producers
- Consumers
- Decomposers

Abiotic Factors

All nonliving things in ecosystem make abiotic factors.

- Temperature
- Moisture
- Light
- Soil
- Air
- Water
- Fire

Natural and artificial resources

Natural Resources:

- Natural resources are elements and materials found in the environment that are valuable to humans for various purposes.

- They are derived from the Earth's natural processes and include categories like minerals, water, forests, air, wildlife, and fossil fuels.
- Natural resources are essential for human survival and economic development, supporting industries such as agriculture, energy, and manufacturing.
- They can be classified into renewable resources (e.g., solar energy, wind energy) and non-renewable resources (e.g., oil, coal) based on their ability to replenish over time.

Artificial Resources:

- Artificial resources, also known as man-made or synthetic resources, are products created by humans through technological processes.
- They include items like manufactured goods, electronics, synthetic materials (plastics, synthetic fibers), and infrastructure (roads, buildings).
- Artificial resources are often derived from natural resources but transformed into new forms with added value through human intervention.
- The production and utilization of artificial resources can have significant environmental and sustainability implications.

Zoology Legends

FOOD CHAINS & its Importance

It is the sequence of organisms in an ecosystem where energy moves from one organism to the next organism. It tells us the sequence to eat and being eaten.

Food chain is important:

1) To understand the feeding relationships between organisms.
2) To explore the energy flow mechanisms and matte circulation.
3) To understand the problems of biomagnification.

Food chains are mainly of two types: grazing food chain and detritus food chain.

Factors Affecting Reproductive Potential

- Number of offspring produced
- Survival to reproductive age
- Reproductive period duration
- Length of time taken to maturity

Advantages of asexual reproduction

What is the effect of pollution on ecosystem

Biodiversity Impact:

Pollution disrupts habitats and food chains, causing a decline in biodiversity. Sensitive species can be harmed or driven to extinction due to polluted environments.

Water Contamination:

Industrial, agricultural, and urban pollutants contaminate water bodies. Toxic chemicals and excess nutrients harm aquatic life and disturb ecosystem balance.

Air Quality Degradation:

Emissions from industries and vehicles release harmful pollutants into the air. Poor air quality affects both human health and the well-being of plants and animals.

Habitat Disruption:

Pollution contributes to the degradation and destruction of natural habitats. Species' survival is compromised, leading to potential extinctions and ecosystem imbalance.

Climate Change Connection:

Pollutants like greenhouse gases contribute to climate change. Altered weather patterns and temperatures disrupt ecosystems and species distribution.

Photoautotrophs and chemoautotrophs

Photoautotrophs use sunlight as energy source for the synthesis of food from inorganic compounds. They include green plants, algae and photosynthetic bacteria. Photosynthesis transform sunlight into chemical energy. Out of total sunlight reaching to earth 30% reflects back, 20% is absorbed by atmosphere, 50% absorbed by earth and less than 1% of sunlight is used by photosynthesis.

Chemoautotrophs obtain energy by oxidizing inorganic compounds. They usually live in extreme environments. e.g., sulfur bacteria.

Why are estuaries important to our environment?

Estuaries are nurseries for crustaceans, molluscs and marine fishes. They are the feeding grounds for wading and swimming birds. Human have drained and filled estuaries. Estuaries are polluted by oil exploration and drilling. Acids rain have also threaten the productivity of estuaries.

Oogenesis

The process of sperms production is called **spermatogenesis** while the process of eggs production is called **oogenesis**.

ZOOLOGY
LEGENDS

Zoology Legends

Muhammad Zaman

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Remember me in you Precious Prayer

Jazak Allah